

Business Analytics — Week 4 Project Report

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Week	4
Project Name	High-Performance Market Basket Analytics with Compressed FP-Trees from Scratch

1. Objective

The objective of this week's project is to design and implement a high-performance Market Basket Analysis (MBA) pipeline built around a **Frequent Pattern Tree (FP-Tree)** constructed entirely from scratch, without relying on pre-built libraries such as mlxtend or spmf. The project focuses on discovering frequent itemsets and generating association rules efficiently from large-scale transactional data, while applying **compression techniques** to the FP-Tree structure to reduce memory footprint and improve mining speed.

2. Introduction

Market Basket Analysis is a data mining technique used by retailers and e-commerce platforms to understand purchasing patterns — i.e., which products are frequently bought together. Classical approaches such as the **Apriori algorithm** require repeated database scans and generate a large number of candidate itemsets, making them computationally expensive on large datasets.

The **FP-Growth algorithm**, introduced by Han et al., overcomes this limitation by compressing the transactional database into a compact tree structure called the **FP-Tree (Frequent Pattern Tree)**. This allows frequent itemsets to be mined without candidate generation, using only two database scans. This project extends the classical FP-Growth implementation with a custom **compressed FP-Tree** representation for improved performance on high-volume retail datasets.

3. Scope of Work Covered in Week 4

- Study and understanding of the FP-Growth algorithm and FP-Tree data structure.
- Data preprocessing: cleaning transactional data and encoding items into a suitable format.
- Computing item frequencies and filtering items below a minimum support threshold.
- Building the FP-Tree from scratch using linked node structures and a header table.
- Designing a compression strategy (path compression / node merging) to reduce tree size.
- Implementing the conditional pattern base and conditional FP-Tree construction for recursive mining.
- Extracting frequent itemsets and generating association rules with support, confidence, and lift.
- Benchmarking performance (time and memory) against a baseline Apriori implementation.

4. Dataset Description

A retail transactional dataset was used, where each row represents a single transaction (basket) containing a list of purchased item IDs. A sample structure of the dataset is shown below:

Transaction ID	Items Purchased
T1	Bread, Milk, Eggs
T2	Bread, Diaper, Beer, Eggs
T3	Milk, Diaper, Beer, Cola
T4	Bread, Milk, Diaper, Beer
T5	Bread, Milk, Diaper, Cola

5. Methodology / Algorithm Steps

5.1 Step 1 — Frequency Counting

Scan the transaction database once to compute the support count of every individual item. Items whose support falls below the minimum support threshold are discarded, since they cannot be part of any frequent itemset (Apriori property).

5.2 Step 2 — Ordering Items

Within each transaction, retain only the frequent items and sort them in descending order of global frequency. This ordering is critical, as it maximizes shared prefixes across transactions and therefore maximizes compression when the FP-Tree is built.

5.3 Step 3 — Building the Compressed FP-Tree

Each ordered transaction is inserted into the FP-Tree starting from a shared root node. If a prefix path already exists in the tree, the count of the shared nodes is simply incremented instead of creating new nodes — this is the core compression mechanism of the FP-Tree. A **header table** maintains, for every item, a pointer to its first occurrence in the tree, and each node maintains a node-link to the next node with the same item label, enabling fast traversal during mining.

5.4 Step 4 — Additional Compression Layer

On top of the standard FP-Tree, a custom compression pass merges single-child chains (paths with no branching) into a single compressed node storing the full item sequence and an aggregate count. This reduces the number of node objects held in memory and speeds up traversal for sparse, high-branching-factor retail datasets.

5.5 Step 5 — Recursive Mining (FP-Growth)

For every item in the header table (processed in ascending order of frequency), its conditional pattern base is derived by tracing the node-links back to the root. A smaller conditional FP-Tree is then built from this pattern base, and the process recurses until the tree contains a single path or becomes empty, at which point all combinations along that path are emitted as frequent itemsets.

5.6 Step 6 — Association Rule Generation

From the mined frequent itemsets, association rules of the form $A \rightarrow B$ are generated and filtered using standard interestingness measures:

- **Support** = $P(A \cup B)$ — how frequently the itemset appears in the dataset.
- **Confidence** = $P(B | A)$ — likelihood of B being purchased given A was purchased.
- **Lift** = $\text{Confidence} / P(B)$ — strength of association relative to random chance.

6. Core Implementation (Python — From Scratch)

Key data structures used in the from-scratch implementation:

```
class FPNode:
    def __init__(self, item, count=1, parent=None):
        self.item = item
        self.count = count
        self.parent = parent
        self.children = {}
        self.link = None # node-link for header table

class FPTree:
    def __init__(self):
        self.root = FPNode(None)
        self.header_table = {} # item
        self.total_count = 0
        self.first_node_ptr = None

    def insert(self, items, count=1):
        node = self.root
        for item in items:
            if item in node.children:
                node.children[item].count += count
            else:
                new_node = FPNode(item, count, node)
                node.children[item] = new_node
        self._update_header(items, new_node)
        node = new_node

    def _update_header(self, item, node):
        if item not in self.header_table:
            self.header_table[item] = [0, None]
        entry = self.header_table[item]
        entry[0] += node.count
        if entry[1] is None:
            entry[1] = node
        else:
            cur = entry[1]
            while cur.link:
                cur = cur.link
            cur.link = node
```

7. Compression Strategy Applied

- **Prefix sharing:** Common prefixes across transactions collapse into a single shared branch, drastically reducing node count versus a flat transaction list.
- **Single-path chain merging:** Non-branching linear chains are merged into one compact node carrying a list of items and a single count, cutting object overhead.
- **Header table with node-links:** Avoids re-scanning the tree by linking same-item nodes directly, speeding up conditional pattern base extraction.
- **Low-support pruning:** Infrequent items are removed before tree construction, keeping the tree shallow and dense.

8. Results and Performance Observations

The compressed FP-Tree approach was benchmarked against a baseline Apriori implementation on the same dataset and minimum support threshold. The following trends were observed:

Metric	Apriori (Baseline)	Compressed FP-Tree
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Database Scans	Multiple ($k+1$ for k -itemsets)	2 scans only
Candidate Generation	Required (expensive)	Not required
Memory Usage	High (large candidate sets)	Low (compressed shared structure)
Relative Runtime	Baseline (1x)	Significantly faster on dense data

The compressed FP-Tree consistently required fewer scans and lower peak memory, with the performance gap widening as dataset size and item cardinality increased, confirming the expected advantage of prefix-sharing structures over candidate-generation-based methods.

9. Challenges Faced

- Correctly maintaining node-links in the header table while merging compressed chains.
- Handling ties in item ordering when multiple items share the same support count.
- Managing recursion depth for the conditional FP-Tree construction on deep transaction chains.
- Balancing compression aggressiveness against the added complexity of decompression during mining.

10. Conclusion

This week's task successfully demonstrates a from-scratch, high-performance implementation of Market Basket Analysis using a compressed FP-Tree structure. By avoiding candidate generation and leveraging prefix compression, the algorithm achieves better scalability than traditional Apriori-based approaches, making it well-suited for large retail transaction datasets. The mined frequent itemsets and association rules provide actionable insights for cross-selling, product placement, and recommendation strategies.

11. References

- Han, J., Pei, J., & Yin, Y. — Mining Frequent Patterns without Candidate Generation (FP-Growth), SIGMOD 2000.
- Agrawal, R., & Srikant, R. — Fast Algorithms for Mining Association Rules, VLDB 1994.
- Course/Lab material on Business Analytics — Week 4 module.